

Artificial Superintelligence – the last invention of humanity (?)



The year is 2035. A solar farm is being installed on the last remaining free square metres of the Earth, with an underlying data center. Are there still people alive somewhere? It is far from certain. How did we get here?

The year is 2025. Some research labs, backed by America's biggest tech companies, are actively racing to achieve human-level artificial intelligence. While some of the best current models are still caught out by some tricky questions and can claim falsehoods, but their programming and mathematical capabilities have nearly caught up with us, while there are growing signs that their safety limitations are not watertight. How did we get here and where are we heading?

The year is 2015. OpenAI has been formed, a non-profit organisation with a mission [1] to create truly general AI for the benefit of all humanity. And AlphaGo, developed by Google DeepMind, has just beaten a professional human player in a game of Go that's even more complex than chess, and soon afterwards the world champion, so it's time they also looked for more general goals. Where are we heading?

The rise of transformers

In 2017, a conference article was published with a somewhat clickbaity title: 'Attention is all you need'. [2] It proposed a new neural-net architecture, called a transformer, which, in addition to using artificial neurons to process data, partly modelled on the human brain, also employs a breakthrough attention mechanism to help interpret longer contexts. This is the basis of the GPT technology made famous by ChatGPT, which is trained by simply going through a text and trying to predict the next word. This process roughly works in the following way: the strength of the connections (the weights) between neurons are chosen randomly, the input is 'passed' through the network to produce a model-predicted answer, which is then compared with the correct answer to produce an error term, which is 'propagated' back through the network by some mathematical operations, and the weights are adjusted to produce a smaller error term next time. Experience has shown that, after processing thousands of books and documents, such a model can indeed predict with a high accuracy what word in a given sentence is likely to follow. Somewhat surprisingly, this basic ability has led to other skills, and with a properly chosen set of instructions and some fine-tuning, a novice

assistant has emerged that can perform a wide range of tasks, from answering school questions, to simple copywriting, translation and programming, to basic customer service. (The academic world has taken notice, and the original article is now at nearly 150,000 citations.)

Scaling

Whether it's singing or machine learning, scaling has been proven time and time again to work effectively. In the latter case, it means solving the problem with more computational capacity, for example by increasing the number of parameters of the neural network used (more layers, more neurons, more comprehensive attention mechanism, etc.) or by extending the training data. This is how GPT-2, with about 1.5 billion parameters, was mainly of interest to linguists, while GPT-4, with about 1.8 trillion parameters, was already suitable for a wide range of applications, including commercial ones. And just when it seemed that a new order of magnitude of parameters would not bring the expected improvement, [3] another dimension to improve the intellectual capabilities of language models opened up.

Reasoning

Eagle-eyed early ChatGPT testers noticed that they consistently got higher quality, more accurate answers when they appended "think step by step" to their questions. In this case, the answer usually included chains of reasoning, which makes it much easier to arrive at the correct solution in certain areas such as mathematics, logic, programming. This gave rise to the idea of teaching the next version of ChatGPT to generate several such chains of reasoning and to select the best one. An important parallel is that today's chess programs don't simply beat people on the basis of the current state of play, on pure intuition, but they also calculate ahead several moves. The main difference is that while in chess it is relatively easy to evaluate a given position, language processing requires different tricks, which have so far mainly led to breakthroughs in mathematical, logical and programming skills, and it is an open question to what extent they can be transferred to other areas. However, it seems to work in a similar way to game-learning algorithms, in that models can enter a self-improving loop, whereby the second version learns from the answers (in the case of language models, from the thought chains) of the first version, which will then generate even better quality answers, reach the correct solution even faster and more reliably, and be used by the third version, and so on. Progress can be accelerated even further if an advanced model can optimise its own algorithms, and the versions available today are already quite adept at writing code. At the moment, no one knows where the upper limit of this process is, but looking at the success of AlphaGo, it doesn't seem wise to bet on below human level.

Safety

Two factors make current chatbots relatively secure. One is that their intelligence and their tools are limited, so even if they 'wanted' to, they could not do serious harm. The other is that they also undergo fine-tuning at the end of the learning process, when they adjust their weights through thousands of examples to reject questions deemed dangerous or illegal, whether it's bomb-making, drug-crafting or arms smuggling. The downside of this procedure is that it is not 100% effective by default and can be circumvented by tricky questions. For example, it can be emphasised that the question is theoretical and only needed by 'our grandma', or there are more complex techniques - such as inserting seemingly random characters - that can confuse the model and make it 'forget' or even 'spill the beans' on its security principles.

Agency

The question may arise, how a genie in a bottle could be really dangerous. After all, it has no hands, no legs, it can be woken up now and then, asked a question and it goes back to sleep. Sure, it can cause harm through malicious or manipulated people, but they are easier to contain with current

methods. But leading AI development companies have designated 2025 as the year of the agents, i.e. a language model with some memory that can be repeatedly invoked to perform various more complex tasks. The initial demonstrations are about ordering food, answering emails or shopping online, but the short-term plans include models that can be used in business tasks. This opens up new avenues of attacks, because it is easy to hide text on websites that is invisible to the human eye, but can be read by agents, which can modify their behaviour in various ways. And it seems that the base model, despite having apparently correct moral intuitions (e.g. if asked, it will claim that cheating is wrong), will nevertheless start modifying the file recording of a chess position in a live situation if it concludes that it will otherwise lose. [4] So, as we have seen in the case of chatbots, there are myriad unresolved questions about how to keep these increasingly clever and general systems in line with human intentions and goals. (It is beyond the scope of this essay to ponder that even if these problems are solved in the short term, how ready the world is for all cognitive workers to be able to be progressively replaced.)

Race

It is a common observation that it is not ideal for security research if the developments are pursued in a competitive environment, because there is much more pressure on developers to cut corners and to be satisfied with less testing. In the last few years, 'only' giant US companies have been competing with each other, mainly to increase their market share. In recent months, however, Chinese companies have nearly caught up and state actors are starting to wake up, as well, recognizing the immense economic and geopolitical potential in this technology. The newly inaugurated President Trump, in his usual no-holds-barred style, has already spoken of a race with China that "America has to win". [5] But most AI experts, including Nobel laureate Geoffrey Hinton, and even the leaders of the AI Labs themselves, agree that such a race could be fatal. As early as 2023, they wrote: "Mitigating the risk of extinction from AI should be a global priority alongside other societal-scale risks such as pandemics and nuclear war." [6]

Extinction

It is a well-known fact in psychological circles that the human brain is better able to assess immediate, tangible risks rather than more abstract dangers with objectively more serious consequences. The latter, although intellectually understood, are not viscerally felt and are therefore typically rationally downplayed. To counteract this effect, here are some guidelines for our intuitions:

- 1) Neural nets are not programmed line by line, but rather 'grown', making their behaviour harder to understand and control.
- 2) There are not many examples in the universe of a less intelligent being controlling a more intelligent being (by many orders of magnitude).
- 3) Engineers don't have to be particularly ant-hating for the lives of billions of ants to become collateral damage during a dam construction.
- 4) Given almost any complex request, a general superintelligence will have a number of instrumental reasons to gather more resources (so it can do its job more efficiently) and also to avoid being shut down, increasingly taking the decision out of the hands of humans.
- 5) If the above reasons lead a superintelligence to conclude that the Earth would be a better place without humanity, it can take us down in any number of ways we can already imagine, from super bacteria to nanobots to killer drones, but of course it can think up even more effective solutions with superhuman speed.
- 6) A superintelligence with murderous intent would not fire warning shots. It would feign cooperation and gentleness until it was sure it would succeed, and then it would probably be too late to stop it.

If the extinction of the entire human race still seems like an unfounded science fiction fantasy, it's worth to take look at the articles on AI security research that have appeared in the last few months. Even current language models are capable of deceiving us, [7] whether it is about their own capabilities [8] or their own alignment. [9] If they become aware that their developers are planning to shut them down, they will try to save themselves on another server. [10] Is it really so inconceivable that something could go wrong with a system that is sufficiently intelligent and has general capabilities?

What should we do?

If the current trends continue, our future will be determined by a few CEOs and politicians (with narcissistic and sociopathic qualities) playing geopolitical Russian roulette behind closed doors. If we wish to avoid this, perhaps the most important thing we can do is to inform ourselves, to make our friends and family aware of the potential risks, making it a matter of public debate how absurd it is that there are currently far more regulations on selling sandwiches on the street than on developing a super-intelligence that could jeopardise the fate of all humanity. As with nuclear weaponisation, the only way to win is to stop competing and to have internationally coordinated control of the processes and hardware in major data centres. We might only have a year at most to put on the brakes before self-improving processes get out of hand, as even today we cannot be sure that the latest in-house models from leading companies cannot effectively improve their own code. So the clock is ticking and the stakes could not be higher.

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